

# 1. Subatomic Properties & Nuclear Species

- **Charge-to-Mass Ratio of Electron ( $e/m_e$ ):**

$$\frac{e}{m_e} \approx 1.75882 \times 10^{11} \text{ C kg}^{-1}$$

- **Masses of Subatomic Particles:**

$$m_e \approx 9.10938 \times 10^{-31} \text{ kg} \approx 5.4858 \times 10^{-4} \text{ u}$$

$$m_p \approx 1.67262 \times 10^{-27} \text{ kg} \approx 1.00727 \text{ u}$$

$$m_n \approx 1.67493 \times 10^{-27} \text{ kg} \approx 1.00866 \text{ u}$$

Relative mass relationship:  $m_n > m_p \gg m_e$  (where  $m_p \approx 1836 m_e$ ).

- **Nuclear Composition & Charge Relationships:**

$$\begin{aligned} \text{Mass Number } (A) &= \text{Atomic Number } (Z) + \text{Neutrons } (N) \\ \implies N &= A - Z \end{aligned}$$

For any chemical species represented as  ${}_Z^AX^{q\pm}$ :

- Protons ( $p^+$ ) =  $Z$
- Neutrons ( $n^0$ ) =  $A - Z$
- Electrons ( $e^-$ ) =  $Z - q$  (for cations with charge  $+q$ )
- Electrons ( $e^-$ ) =  $Z + q$  (for anions with charge  $-q$ )

- **Nuclear Species Classifications:**

- **Isotopes:** Same  $Z$ , different  $A$  (e.g.,  ${}^{12}_6\text{C}$  and  ${}^{14}_6\text{C}$ ).
- **Isobars:** Same  $A$ , different  $Z$  (e.g.,  ${}^{40}_{18}\text{Ar}$  and  ${}^{40}_{20}\text{Ca}$ ).
- **Isotones:** Same  $N = A - Z$  (e.g.,  ${}^{14}_6\text{C}$  and  ${}^{15}_7\text{N}$ ).
- **Isodiaphers:** Same  $(N - Z) = A - 2Z$ .
- **Isoelectronic Species:** Same total electron count ( $e^-$ ).

- **Nuclear Dimensions:**

$$R = R_0 A^{1/3} \quad (R_0 \approx 1.2 \times 10^{-15} \text{ m}) \implies V \propto A$$

# 2. Wave Nature & Photoelectric Solutions

- **Electromagnetic Radiation Relationships:**

$$c = \nu \lambda \implies \lambda = \frac{c}{\nu}, \quad \nu = \frac{c}{\lambda}$$

where speed of light in vacuum  $c \approx 2.998 \times 10^8 \text{ m s}^{-1}$ .

- **Wavenumber Definition:**

$$\bar{\nu} = \frac{1}{\lambda} = \frac{\nu}{c} \quad (\text{expressed typically in } \text{m}^{-1} \text{ or } \text{cm}^{-1})$$

- **Planck's Quantum Energy Equation:**

$$E = h\nu = \frac{hc}{\lambda} = hc\bar{\nu}$$

Planck's constant  $h \approx 6.626 \times 10^{-34} \text{ J s} \approx 4.136 \times 10^{-15} \text{ eV s}$ .

Handy product values:  $hc \approx 1.986 \times 10^{-25} \text{ J m} \approx 1240 \text{ eV nm}$ .

- **Electromagnetic Spectrum Energy Gradient:**

Radio < Microwave < Infrared < Visible < UV < X-ray < Gamma ray

- **Einstein's Photoelectric Equation:**

$$E_{\text{photon}} = W_0 + K.E._{\text{max}} \implies h\nu = h\nu_0 + K.E._{\text{max}}$$

where work function  $W_0 = h\nu_0$  (threshold energy),  $\nu_0$  is the threshold frequency, and maximum kinetic energy is:

$$K.E._{\text{max}} = \frac{1}{2}m_e v^2 = h(\nu - \nu_0) = hc \left( \frac{1}{\lambda} - \frac{1}{\lambda_0} \right)$$

- **Threshold Wavelength ( $\lambda_0$ ):**

$$\lambda_0 = \frac{hc}{W_0}$$

- **Photon Emission Rate from Power Source ( $P$  in Watts):**

$$N_{\text{photons/sec}} = \frac{\text{Power } (P)}{E_{\text{photon}}} = \frac{P\lambda}{hc}$$

- **Photoelectric Proportionality Principles:**

Light Intensity ( $I$ )  $\propto$  Number of emitted photoelectrons per second

Light Frequency ( $\nu$ )  $\propto$  Kinetic energy of emitted electrons ( $K.E._{\text{max}}$ )

### 3. Bohr's Quantized Hydrogen Model

- **Quantization of Angular Momentum:**

$$m_e v r = n \frac{h}{2\pi} = n \hbar \quad (n \in \mathbb{Z}^+)$$

- **Hydrogenic Orbit Radii ( $r_n$ ):**

$$r_n = a_0 \frac{n^2}{Z} \approx 52.9 \frac{n^2}{Z} \text{ pm} = 0.529 \frac{n^2}{Z} \text{ \AA}$$

where  $a_0 = 52.9 \text{ pm}$  is the first Bohr radius of Hydrogen ( $Z = 1$ ).

- **Hydrogenic Energy Levels ( $E_n$ ):**

$$\begin{aligned} E_n &= -E_H \frac{Z^2}{n^2} \approx -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J} \\ &= -13.6 \frac{Z^2}{n^2} \text{ eV} = -1312 \frac{Z^2}{n^2} \text{ kJ mol}^{-1} \end{aligned}$$

where  $E_H = 13.6 \text{ eV}$  is the ionization energy of ground-state Hydrogen.

- **Orbital Electron Velocity ( $v_n$ ):**

$$v_n = v_1 \frac{Z}{n} \approx 2.188 \times 10^6 \frac{Z}{n} \text{ m s}^{-1}$$

where  $v_1$  is the first orbit orbital electron speed of Hydrogen.

- **Ionisation Energy ( $I.E.$ ):**

$$I.E. = E_\infty - E_{\text{ground}} = 0 - E_1 = 13.6 Z^2 \text{ eV}$$

- **Consecutive Orbit Energy Gap Trend:**

$$\Delta E = E_{n+1} - E_n \implies \text{As } n \uparrow, \quad \Delta E \downarrow$$

### 4. Rydberg-Ritz Spectral Transitions

- **Transition Wavenumber Calculation:**

$$\bar{\nu} = \frac{1}{\lambda} = R_H Z^2 \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Rydberg constant  $R_H \approx 109,677 \text{ cm}^{-1} \approx 1.097 \times 10^7 \text{ m}^{-1}$ .

- **Transition Energy ( $\Delta E$ ):**

$$\Delta E = h\nu = hc\bar{\nu} = E_H Z^2 \left( \frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

- **Wavelength Boundaries:**

$$\text{Shortest Wavelength (Series Limit): } n_2 = \infty \implies \lambda_{\min} = \frac{n_1^2}{R_H Z^2} \propto \frac{1}{Z^2}$$

$$\text{Longest Wavelength (First Line): } n_2 = n_1 + 1 \implies \lambda_{\max} = \frac{n_1^2 (n_1 + 1)^2}{(2n_1 + 1) R_H Z^2}$$

- **Total Spectral Emission Lines ( $N_{\text{lines}}$ ):** During a transition from higher orbit  $n_2$  to lower orbit  $n_1$ :

$$N_{\text{lines}} = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

If the transition terminates at the ground state ( $n_1 = 1$ ):

$$N_{\text{lines}} = \frac{n_2(n_2 - 1)}{2}$$

- **Hydrogen Spectral Series Regions:**

| Series   | Lower State ( $n_1$ ) | Upper State ( $n_2$ ) | Spectral Region         |
|----------|-----------------------|-----------------------|-------------------------|
| Lyman    | 1                     | 2, 3, 4, ...          | Ultraviolet (UV)        |
| Balmer   | 2                     | 3, 4, 5, ...          | Visible                 |
| Paschen  | 3                     | 4, 5, 6, ...          | Near-Infrared (Near-IR) |
| Brackett | 4                     | 5, 6, 7, ...          | Mid-Infrared (Mid-IR)   |
| Pfund    | 5                     | 6, 7, 8, ...          | Far-Infrared (Far-IR)   |

### 5. de Broglie Matter Waves & Uncertainty Limit

- **de Broglie Matter Wave Equation:**

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

- **Wavelength and Kinetic Energy Relation:**

$$K.E. = \frac{p^2}{2m} \implies p = \sqrt{2m \cdot K.E.} \implies \lambda = \frac{h}{\sqrt{2m \cdot K.E.}}$$

- **Potential Accelerated Electron Wavelength:**

$$\lambda = \frac{h}{\sqrt{2m_e eV}} \approx \frac{1.227}{\sqrt{V}} \text{ nm} = \frac{12.27}{\sqrt{V}} \text{ \AA}$$

where  $V$  is the potential difference in Volts.

- **Heisenberg's Uncertainty Principle:**

$$\Delta x \cdot \Delta p_x \geq \frac{h}{4\pi} \implies \Delta x \cdot (m\Delta v_x) \geq \frac{h}{4\pi}$$

- **Numerically Equal Uncertainties** ( $\Delta x = \Delta v_x$ ):

$$(\Delta v_x)^2 \geq \frac{h}{4\pi m} \implies \Delta v_{x,\min} = \sqrt{\frac{h}{4\pi m}}$$

## 6. Quantum Mechanical Model & Orbitals

- **Schrödinger Wave Equation (Time-Independent):**

$$\hat{H}\psi = E\psi \implies \nabla^2\psi + \frac{8\pi^2m}{h^2}(E - V)\psi = 0$$

where  $|\psi|^2$  represents the probability density of finding the electron at a point.

- **Radial and Angular Nodes:**

$$\text{Radial Nodes (Spherical)} = n - l - 1$$

$$\text{Angular Nodes (Nodal Planes)} = l$$

$$\text{Total Nodes} = n - 1$$

- **Quantum Coordinates & Selection Rules:**

- **Principal ( $n$ ):**  $n \in \{1, 2, 3, \dots\}$  (Determines size and energy of shell).
- **Azimuthal ( $l$ ):**  $0 \leq l \leq n-1$  (Determines orbital shape / angular momentum).
- **Magnetic ( $m_l$ ):**  $-l \leq m_l \leq +l$  (Determines orbital orientation).
- **Spin ( $m_s$ ):**  $m_s \in \{-\frac{1}{2}, +\frac{1}{2}\}$  (Determines spin direction).

- **Shell and Subshell Boundaries:**

$$\text{Number of Orbitals in Subshell} = 2l + 1$$

$$\text{Maximum Electrons in Subshell} = 2(2l + 1)$$

$$\text{Number of Orbitals in Shell} = n^2$$

$$\text{Maximum Electrons in Shell} = 2n^2$$

- **Orbital Shapes:**

- $s$  orbital ( $l = 0$ ): Spherical symmetric.
- $p$  orbital ( $l = 1$ ): Dumb-bell (three orientations:  $p_x, p_y, p_z$ ).
- $d$  orbital ( $l = 2$ ): Cloverleaf / double dumb-bell ( $d_{xy}, d_{yz}, d_{xz}, d_{x^2-y^2}, d_{z^2}$ ).

- **Orbital Angular Momentum ( $L$ ):**

$$L = \sqrt{l(l+1)} \frac{h}{2\pi} = \sqrt{l(l+1)} \hbar$$

- **Spin-Only Magnetic Moment ( $\mu_s$ ):**

$$\mu_s = \sqrt{n(n+2)} \text{ Bohr Magnetons (B.M.)}$$

where  $n$  is the total count of unpaired electrons.

## 7. Electronic Configurations & Filling Rules

- **Aufbau Principle (The  $n + l$  Rule):** Electrons occupy subshells in order of increasing energy.
  - Energy increases as the sum ( $n + l$ ) increases.
  - For equivalent ( $n + l$ ) sums, the subshell with the lower  $n$  value fills first.
- **Pauli Exclusion Principle:** No two electrons in a single atom can share the identical set of all four quantum numbers ( $n, l, m_l, m_s$ ).
- **Hund's Rule of Maximum Multiplicity:** Degenerate orbitals of a subshell must be singly occupied with parallel electron spins before pairing occurs.
- **Extra Stability of Half and Fully-Filled Subshells:** Symmetrical configuration charge distributions minimize inter-electronic repulsion, and high exchange energy stabilizes configurations such as:

$$\text{Chromium (} Z = 24) \implies [\text{Ar}]3d^54s^1 \quad (\text{not } 3d^44s^2)$$

$$\text{Copper (} Z = 29) \implies [\text{Ar}]3d^{10}4s^1 \quad (\text{not } 3d^94s^2)$$

Exchange energy stabilization count ( $E_{\text{ex}}$ ) for  $k$  electrons with parallel spin:

$$E_{\text{ex}} = \frac{k(k-1)}{2}$$